Bot Hunter Analysis Report

1. Problem Statement

Bot Hunter analyses fictitious ad-click traffic to identify events that look more like automated bot activity than legitimate user behaviour. The input is an unlabelled click log, so the task is not to prove fraud event by event. The task is to find defensible anomaly patterns, explain the evidence behind them, and produce a binary submission.tsv prediction where false positives and false negatives are treated as roughly equal in cost.

This run read data/bot-hunter-dataset.tsv and analysed 149,239 click events. Each row contains an event identifier, timestamp, region, browser, operating system, and click URL. The URL query string carries behaviour signals such as clicked domain (d), search query (q), time to click (ttc), and country-like context (ct).

This report should be read as evidence for review, not as proof of fraud for every individual event. The dataset does not include ground-truth labels, so the system cannot claim measured precision, measured recall, or a calibrated fraud probability.

The submitted output is submission.tsv. It keeps the required binary is_bot decision and adds operational_tier so the same prediction can be used carefully in business workflows.

2. Methodology And Rationale

Bot Hunter uses two classifiers because each covers a different risk. The rules classifier catches patterns that are easy to explain and audit, such as repeated query/domain pairs, mechanical timing, and dense bursts. The Extended Isolation Forest model catches unusual combinations across engineered features that fixed rules may miss.

Component	Current choice	Why this choice was made
Rules classifier	Weighted heuristic rules	Transparent, auditable, and suitable for explaining repeated or mechanical click patterns.
ML classifier	Extended Isolation Forest model	Works without labels and can detect unusual combinations across engineered features.
Score blend	0.58 * heuristic_score + 0.42 * ml_score	Keeps explainable rule

evidence dominant while still using the anomaly model. |
| Main cutoff | 97.5th percentile combined-score threshold | Keeps the selected population stable for an unlabelled batch. |
| Override | heuristic_score >= 0.62 | Protects high-confidence rule evidence from being missed by model ranking shifts. |

The final score is a weighted blend:

text
combined_score = (0.58 * heuristic_score) + (0.42 * ml_score)

The rules layer is weighted slightly higher because it is easier to explain and audit. The anomaly model still matters because it can identify unusual combinations that a small rule set may not capture.

Raw URLs are not model-ready. Bot Hunter parses each query string and converts it into behavioural features. Count features are log-transformed because click-log counts are heavy-tailed: a few domains or queries may appear many times while most appear once. kp and sld are treated as categorical identifiers, not ordered numeric measurements. Their raw values remain in artifacts/features.tsv for audit, but the model uses log_kp_count and log_sld_count to learn concentration without inventing false numeric distance between categories.

The rules layer currently scores:

- repeated query/domain pairs
- repeated queries
- confirmed query repetition
- high-volume clicked domains
- dense region/browser/OS clusters
- exact time-to-click reuse
- same-second bursts
- dense burst repetition clusters
- concentrated ct context paired with repetition and clustering
- implausibly fast clicks
- moderately long or extreme time-to-click values
- regular pseudo-session inter-arrival timing

Rule contributions are separated into strong and supporting evidence. Strong rules represent direct mechanical or replay-like patterns, such as impossible timing or repeated query/domain behaviour. Supporting rules provide context, such as volume, concentration, or broad clustering. Supporting contributions are capped so several weak signals cannot add up to the same meaning as a strong bot signal on their own.

Rule strength	Meaning	Score treatment
Strong	Direct mechanical or replay evidence	Contributes its full rule weight
Supporting	Contextual or weaker evidence	Capped together at 0.24

The exact time-to-click reuse rule uses a 99th-percentile reuse-count threshold with an absolute floor. The main repetition and concentration rules also use 99th-percentile batch thresholds with the previous fixed count and total-rate cutoffs kept as guardrails. This lets the rules adapt to the observed population while avoiding weak duplicate counts in small runs.

The ct rule, confirmed query repetition, and dense burst repetition rules are conjunctive. In practical terms, they require multiple signals to be present before adding score. This is safer than treating broad context, such as country-like concentration, as a standalone bot indicator.

The anomaly classifier uses the engineered feature matrix. The main feature families are:

- region/browser/OS frequency
- global ct country frequency
- sub-200 ms click flags
- local burst density
- query entropy
- query, domain, and query/domain repetition counts
- same-second and exact time-to-click reuse counts
- timing magnitude after log transformation
- low-cardinality kp and sld value frequencies

High-volume domain frequency and global country frequency are down-weighted to 0.50 and 0.50, respectively. The low-cardinality kp and sld frequency features are down-weighted to 0.50 and 0.25. Events isolated quickly by random hyperplane splits receive higher anomaly scores. EIF is the only production anomaly model; alternate ML backends and supervised pilots have been removed.

3. Current Statistical Findings

Bot Hunter selected 3,731 of 149,239 events as likely bot traffic. That is 2.50% of the run.

Metric	Current value	Plain-English meaning
Total events analysed	149,239	Number of click events in the batch.
Selected bot events	3,731	Events marked is_bot = 1.
Selected bot rate	2.50%	Share of the batch selected as likely bot traffic.
Suppress tier	1,863	Strongest operational candidates, subject to policy approval.
Quarantine tier	1,868	Suspicious events that should be reviewed or sampled.
Monitor tier	145,508	Events not selected for bot action.
Combined-score cutoff	0.568053	Run-specific anomaly threshold.
Heuristic-only flag rate	1.36%	Share flagged by rules before model blending.
ML agreement-tail rate	2.50%	Reference tail used to compare ML agreement.
Operational confidence estimate	74.49%	Signal-based estimate, not measured precision.

Top explainable rule patterns from the current run:

Pattern	Events	How to explain it
repeated query	3,403	The same search term appears unusually often.
repeated query/domain pair	2,086	The same search term repeatedly clicks the same domain.
confirmed query repetition	2,023	Query repetition is backed by additional context.
very short query	1,798	Short terms can be legitimate, but are useful supporting evidence.
high-volume clicked domain	1,732	A domain receives unusually concentrated clicks.
heavy region/browser/OS cluster	1,168	Traffic is concentrated in a narrow device/server footprint.
concentrated ct context	904	Country-like context is concentrated with repetition and clustering.
same-second click burst	677	Multiple clicks happen in the same second.
moderately long time-to-click	431	Timing is unusual enough to support other evidence.
dense burst repetition cluster	236	Burstiness and repeated behaviour appear together.

Method agreement from the current run:

Bucket	Events	Interpretation
Heuristic + ML	1,726	Strongest review candidates because both methods agree.
Heuristic only	311	Explainable rule hits that are not extreme in the ML feature space.
ML only	2,005	Multivariate anomalies that need careful sampling or review.
Neither strong	145,197	Traffic not strongly indicated by either method.

Operational anomaly classes from the current run:

Class	Selected events	Recommended handling
Repetition with supporting context	1,877	Review as replay-like traffic. Suppress only when the event is already in the suppress tier; otherwise quarantine or sample.
Compound burst/replay	562	Treat suppress-tier events as strong operational candidates; quarantine the rest for timing-pattern review.
ML-tail multivariate anomaly	509	Quarantine or sample. Do not suppress automatically without feature-deviation review or labels.
Repetition with timing anomaly	355	Prioritise when suppress-tier or when timing evidence is strong; otherwise quarantine for sampling.
Repetition dominated	350	Use as an explainable replay candidate. Quarantine lower-score cases when ML agreement is absent.
Supporting context plus combined tail	77	Monitor or quarantine. These are useful for trend review, not standalone suppression.
Other combined-tail anomaly	1	Sample manually before taking action.

4. Explanation Of Anomalies Found

Repetition is suspicious when the same query, clicked domain, and device-like context appear together. Timing matters because some click patterns are difficult for humans to produce consistently. Country-like ct concentration is supporting evidence only; it should not be treated as a country-blocking rule. ML-only events are unusual, but unusual does not automatically mean fraudulent.

Operational anomaly classes derived from the current unlabelled run; these are not proven fraud labels.

The classes below group the 3,731 selected events from this run.
The grouping is backed by full-run rule contributions, rule strength and family

fields, heuristic and ML scores, method-agreement buckets, operational tiers,
and the run-specific thresholds described later in this report.

Class	Selected events	Data backing	Suggested handling
Repetition with supporting context	1,877	tiers: suppress 1,327, quarantine 550; methods: Heuristic + ML 1,198, Heuristic only 186, Combined tail 493; top rules: repeat_query (1,877), repeat_query_domain (1,384), confirmed_query_repetition (1,384)	Review as replay-like traffic. Suppress only when the event is already in the suppress tier; otherwise quarantine or sample.
Compound burst/replay	562	tiers: suppress 154, quarantine 408; methods: Heuristic + ML 150, Heuristic only 11, Combined tail 401; top rules: repeat_query (558), same_second_burst (441), short_query (285)	Treat suppress-tier events as strong operational candidates; quarantine the rest for timing-pattern review.
ML-tail multivariate anomaly	509	tiers: quarantine 509; methods: ML only 509; This class is grouped by anomaly-model agreement, not by rule explanations. Low-weight rules may be present, but they do not cross the heuristic override threshold.	Quarantine or sample. Do not suppress automatically without feature-deviation review or labels.
Repetition with timing anomaly	355	tiers: suppress 199, quarantine 156; methods: Heuristic + ML 195, Heuristic only 4, Combined tail 156; top rules: repeat_query (355), moderate_long_ttc (283), repeat_query_domain (199)	Prioritise when suppress-tier or when timing evidence is strong; otherwise quarantine for sampling.
Repetition dominated	350	tiers: suppress 183, quarantine 167; methods: Heuristic + ML 183, Heuristic only 110, Combined tail 57; top rules: repeat_query_domain (350), repeat_query (293), confirmed_query_repetition (293)	Use as an explainable replay candidate. Quarantine lower-score cases when ML agreement is absent.
Supporting context plus combined tail	77	tiers: quarantine 77; methods: Combined tail 77; top rules: same_second_burst (74), high_volume_domain (62), heavy_device_cluster (62)	Monitor or quarantine. These are useful for trend review, not standalone suppression.
Other combined-tail anomaly	1	tiers: quarantine 1; methods: Combined tail 1; top rules: same_second_burst (1), fast_click (1)	Sample manually before taking action.

The ML-tail population contains 2,005 events with high anomaly
scores and heuristic scores below the rule override threshold. Only the subset
that also crossed the combined-score cutoff is counted as selected traffic in
the class table. This keeps ML-only anomalies visible without describing them
as rule-derived replay evidence.

Concrete examples from the current run:

- Repetition with supporting context: evt_147679 clicked www.amazon.de for query nomnem;

combined 0.9183, rules 0.8600, ML 0.9988, tier suppress, method Heuristic + ML; rules: repeat_query_domain, repeat_query, confirmed_query_repetition, high_volume_domain, heavy_device_cluster, concentrated_ct_context, short_query.
- Compound burst/replay: evt_097085 clicked duckduckgo.com for query vielle motoneuron; combined 0.9982, rules 1.0000, ML 0.9957, tier suppress, method Heuristic + ML; rules: repeat_query_domain, repeat_query, confirmed_query_repetition, heavy_device_cluster, same_second_burst, dense_burst_repetition_cluster, concentrated_ct_context, moderate_long_ttc.
- ML-tail multivariate anomaly (2,005 events in the wider population): evt_082067 clicked www.firefold.com for query splenocyte; combined 0.7637, rules 0.6000, ML 0.9896, tier quarantine, method ML only.
- Repetition with timing anomaly: evt_004943 clicked www.amazon.ca for query nomnem; combined 0.9184, rules 0.8600, ML 0.9991, tier suppress, method Heuristic + ML; rules: repeat_query_domain, repeat_query, confirmed_query_repetition, high_volume_domain, heavy_device_cluster, extreme_ttc, short_query.
- Repetition dominated: evt_009777 clicked www.overstock.com for query fuzz sulphid; combined 0.7781, rules 0.6200, ML 0.9964, tier suppress, method Heuristic + ML; rules: repeat_query_domain, repeat_query, confirmed_query_repetition.
- Supporting context plus combined tail: evt_089535 clicked www.amazon.co.uk for query aw; combined 0.6109, rules 0.3600, ML 0.9574, tier quarantine, method Combined tail; rules: high_volume_domain, heavy_device_cluster, same_second_burst, moderate_long_ttc, short_query.
- Other combined-tail anomaly: evt_032590 clicked find.gmx.com for query balm thyrohyal youre eight; combined 0.5687, rules 0.3000, ML 0.9396, tier quarantine, method Combined tail; rules: same_second_burst, fast_click.

Practical filtering options for similar unlabelled datasets:

Filter	Use	Caveat
Conservative suppression review: operational_tier == ’suppress’	Start here for the strongest operational candidates. Still requires policy approval because labels are unavailable.	Check policy, billing, and customer-impact rules before action.
Quarantine for manual review: operational_tier == ’quarantine’	Hold, sample, or delay action on suspicious traffic that is not strong enough for direct suppression.	Use sampling to estimate likely false positives before suppression.
Explainable replay review: anomaly_class in repetition_with_supporting_context, compound_burst_replay, repetition_with_timing, repetition_dominated	Focus reviewer time on repeated query/domain behaviour with clear rule evidence.	Validate repeated-pattern assumptions against campaign context.
ML-tail sampling: ml_score >= 0.975 and heuristic_score < 0.62	Sample for future feature-deviation work; do not treat as proven fraud without labels.	Needs feature-

deviation review because rule evidence is below override. |

Use these filters as review controls. suppress is the strongest operational tier, quarantine is the safer default for ambiguous or ML-only traffic, and monitor keeps non-selected traffic available for drift checks and future labels.

5. Recommended Business Actions

Separate prediction from action. is_bot is the compatibility prediction; operational_tier is the safer business-control layer.

Tier	Recommended action	Business assumption
suppress	Exclude from billing or downstream metrics after policy approval.	The organisation accepts limited review risk for high-confidence traffic.
quarantine	Hold, delay, sample, or manually review before suppression.	Ambiguous traffic should not be automatically removed.
monitor	Keep for trend monitoring and future labels.	Non-selected traffic can still help detect drift later.

Operational split in this run:

- suppress: 1,863 events
- quarantine: 1,868 events
- monitor: 145,508 events

Use suppress for the strongest operational candidates after policy approval. Use quarantine as the default action for ambiguous or ML-only traffic. Do not automatically block traffic solely because it is in the ML tail.

Filtering options are in Section 4. They are review controls for similar unlabelled datasets, not ground-truth fraud rules.

6. Probability Perspective

The current operational confidence estimate is 74.49%. This is not measured precision because the dataset has no ground-truth labels.

Confidence is higher when independent signals agree: for example, when a strong repeated query/domain rule hit also lands in the upper anomaly-model tail. Confidence is lower for ML-only events because anomaly detection can also surface legitimate edge cases such as unusual campaigns, regional spikes, testing traffic, or rare but valid user behaviour.

The estimate is based on rule/model agreement, strength of rule evidence, operational tier, selected anomaly class, and the known absence of labels. It should guide review priority, not replace labelled validation or policy approval.

7. Generalisation, Trade-Offs, And Limitations

The approach should generalise to bot traffic that is repetitive, mechanically timed, or concentrated in narrow device and query patterns. It may miss bots that deliberately randomise timing, distribute activity across many query/domain pairs, or closely imitate human behaviour.

The main trade-off is explainability versus coverage. Rules are easier to defend but can miss novel patterns. The anomaly model broadens coverage but needs careful interpretation because it finds unusual events, not necessarily fraudulent events.

Known limitations:

- There are no ground-truth labels, so measured precision, recall, and calibration cannot be reported.
- Legitimate campaigns can create high repetition, dense clusters, or regional concentration.
- The ct rule is supporting evidence only; it should not be used as a country-level blocking rule.
- The regular inter-arrival rule uses pseudo-session groups because there is no explicit user or session identifier.
- The anomaly score is relative to the current batch and should be monitored for drift across future runs.

Material changes in geography, campaign volume, inventory, browser mix, or feature extraction should trigger fresh threshold review.

8. Future Work

The next useful improvements are:

1. Labelled validation from manual review, chargebacks, or confirmed invalid traffic feedback.

2. Feature-deviation explanations for ML-tail events.
3. Historical drift monitoring for flagged rates and score distributions.
4. Campaign-level or inventory-level normalisation if metadata becomes available.
5. Calibrated probabilities once trusted labels exist.
6. A feedback loop from reviewed suppress and quarantine decisions.

Appendix A: Metric Definitions

Metric	Definition
is_bot	Required binary prediction in submission.tsv; 1 means selected as likely bot traffic.
heuristic_score	Score from transparent rules such as repetition, timing, and burst checks.
ml_score	Anomaly score from the Extended Isolation Forest feature model.
combined_score	Weighted blend of rule and ML evidence.
Combined-score cutoff	Run-specific threshold used to select the top anomaly population.
Heuristic override	Rule-score threshold that selects strong explainable rule hits.
Operational tier	Business action layer: suppress, quarantine, or monitor.
Operational confidence estimate	Signal-based estimate from method agreement; not measured precision.
Method bucket	Agreement category showing whether rules, ML, both, or neither were strong.
Anomaly class	Human-readable grouping of selected events by dominant evidence pattern.

Appendix B: Feature Definitions

Feature	Meaning
log_domain_count	Log-scaled frequency of the clicked domain.
log_query_count	Log-scaled frequency of the search query.
log_query_domain_count	Log-scaled frequency of the query/domain pair.
log_device_count	Log-scaled frequency of the region/browser/OS cluster.
log_country_count	Log-scaled frequency of ct, the country-like URL parameter.
log_same_second_count	Log-scaled number of clicks sharing the same second.
log_ttc_count	Log-scaled reuse count for exact time-to-click values.
log_kp_count	Log-scaled frequency of the kp URL parameter value.
log_sld_count	Log-scaled frequency of the sld URL parameter value.
kp	Raw kp URL parameter retained for audit; excluded from ML features.

sld	Raw sld URL parameter retained for audit; excluded from ML features.
hour	Event hour extracted from the timestamp.
log_ttc_seconds	Log-scaled time-to-click duration.
is_sub_200ms_click	Flag for click timing below a plausible human reaction threshold.
log_pseudo_session_10s_click_count	Log-scaled burst density within pseudo-session-style groups.
query_entropy	Text randomness measure for the search query.

Appendix C: Model Definition

Extended Isolation Forest is an unsupervised anomaly-detection model. It does not learn from known bot labels. Instead, it asks which events are easier to isolate from the rest of the batch using random feature-space splits.

In plain English, events that look very different from the main population tend to be isolated more quickly and receive higher anomaly scores. This is useful for an unlabelled dataset, but it also means the model detects unusual behaviour rather than confirmed fraud. The model output is strongest when it agrees with transparent rule evidence and weakest when it stands alone without feature-deviation review.

Appendix D: Thresholds And Decision Logic

An event is selected as a bot when either condition is true:

text
```
combined_score > 0.568053
or heuristic_score >= 0.62
```

The combined-score threshold is the run-specific 97.5th-percentile cutoff. It keeps the selected volume stable for an unlabelled dataset while letting the ML model influence borderline cases.

The heuristic override protects high-confidence, explainable rule hits. For example, a strongly repeated query/domain pattern with mechanical timing should not be missed only because the anomaly ranking moved after a feature change.

Adaptive heuristic thresholds used in this run:

Rule	Computed threshold	Basis

Concentrated ct context (concentrated_ct_context)	29013	99th-percentile threshold, floor 1000, rate guardrail 14923
Heavy region/browser/OS cluster (heavy_device_cluster)	43674	99th-percentile threshold, floor 600, rate guardrail 5223
High-volume clicked domain (high_volume_domain)	2238	99th-percentile threshold, floor 200, rate guardrail 2238
Repeated query (repeat_query)	149	99th-percentile threshold, floor 12, rate guardrail 149
Repeated query/domain pair (repeat_query_domain)	37	99th-percentile threshold, floor 4, rate guardrail 37
Reused exact time-to-click (reused_ttc)	40	99th-percentile threshold, floor 40
Same-second click burst (same_second_burst)	6	99th-percentile threshold, floor 4

In this run:

- heuristic-only flag rate: 1.36%
- ML agreement-tail reference rate: 2.50%
- operational confidence estimate: 74.49%